SERVICENOW PROJECT DOCUMENTATION — PHASE 3

3. Project Design Phase

Automating Employee Data Import in ServiceNow

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Prepared by the ServiceNow Implementation Team

Section 1 — Introduction to Project Design Phase

1. Introduction to Project Design Phase

The Project Design Phase is a pivotal stage within the software development and systems implementation lifecycle that serves as the intellectual bridge between the documented requirements of a project and the actual technical implementation that follows. While the preceding phases establish what the system must accomplish and why it must accomplish it, the design phase addresses the fundamental question of how the system will be built. It is during this phase that abstract requirements are translated into a concrete, actionable technical blueprint that guides all subsequent development activity.

In the context of the Automating Employee Data Import project, the design phase focuses on defining the system architecture, specifying the data flows between components, designing the database structure and table relationships, determining the field-level mapping strategy, planning the Incident form layout customisations, establishing the configuration design for priority management, and mapping the complete process workflow. Each of these design decisions is made in direct response to the requirements identified during the Requirement Analysis phase and is shaped by the capabilities and constraints of the ServiceNow platform.

A well-executed design phase delivers several important benefits to the overall project. It reduces the risk of implementation errors by ensuring that all developers have a clear and unambiguous specification to work from before any configuration begins. It exposes potential issues, such as conflicting field mappings, unsupported reference resolutions, or incorrect inheritance assumptions, at the design stage where they are far less costly to address than when they are discovered during or after implementation. It also provides stakeholders with a tangible artefact that can be reviewed and approved before resource-intensive development work commences, thereby ensuring alignment between the project team and the project's sponsors or evaluators.

This document presents the complete design specification for the Automating Employee Data Import project, organised into nine design sections that collectively define all aspects of the solution's structure and behaviour. Each section is accompanied by supporting tables and descriptive explanations to ensure that the design intent is communicated with clarity and precision, and that the document is suitable for direct use as the authoritative reference during the subsequent implementation phase.

Section 2 — System Architecture Overview

2. System Architecture Overview

The system architecture of the Automating Employee Data Import solution is structured as a layered, sequential pipeline in which data originates from an external source, passes through a series of transformation and storage stages within ServiceNow, and ultimately becomes available for consumption by the platform's Incident Management module. Each layer in the architecture performs a distinct and well-defined role, and the output of each layer serves as the input to the layer that follows.

The architecture is deliberately designed to leverage ServiceNow's native, built-in capabilities, specifically the Import Sets and Transform Maps modules, rather than relying on custom scripting or external integration tools. This approach maximises reliability, reduces implementation complexity, and ensures that the solution remains maintainable by standard ServiceNow administrators without requiring specialised coding expertise.

2.1 Architecture Layer Reference

Layer	System Layer	Component	Role in the Project
L1	**External Data Source**	`Google Sheets / Excel`	Serves as the origin point for all employee data. Data is structured in a tabular format with defined column headers. The file is exported in XLSX or CSV format for compatibility with ServiceNow's Import Sets module.
L2	**Data Staging Layer**	`Import Set Table`	Acts as a temporary staging environment that receives and holds raw imported data before transformation. Provides a safe intermediate layer where data can be inspected and validated before being committed to production tables.
L3	**Transformation Layer**	`Transform Map`	Processes records from the Import Set staging table, applies defined field mappings, resolves reference fields through Dot Walking expressions, and writes transformed records to the designated target table.
L4	**User Data Layer**	`sys_user Table`	The permanent storage layer for all employee user records in ServiceNow. Serves as the authoritative source of employee information for all modules on the platform, including Incident Management.
L5	**Incident Management Layer**	`incident Table`	Consumes employee data from sys_user through reference fields. Displays department, email, and manager information within incident records via Dot Walking, enabling accurate incident routing and escalation.
L6	**Data Traversal Mechanism**	`Dot Walking`	Traverses reference field relationships between tables to retrieve and display related data values automatically within the Incident form, without requiring manual agent input.

Layer	System Layer	Component	Role in the Project
L7	Configuration Layer	`Dictionary Override`	Applied to the Priority field of the Incident table to customise its default value, overriding the inherited system default to align with the organisation's incident management policy.

2.2 Architecture Design Principles

The system architecture is founded on the following five design principles, each of which informed specific architectural decisions throughout the design process:

- Separation of Concerns: Each layer of the architecture performs a single, clearly bounded function, preventing the conflation of responsibilities and simplifying troubleshooting and long-term maintenance.
- Use of Native Platform Capabilities: All components are implemented using ServiceNow's built-in tools and modules, eliminating external dependencies and ensuring compatibility with future platform updates.
- Data Integrity by Design: The use of a staging layer before committing data to production tables ensures that imported records can be validated and corrected before they affect the live sys_user table.
- Repeatability: The architecture supports repeated execution of the import pipeline with different data files, enabling the organisation to refresh employee records on a recurring basis without reconfiguration.
- Referential Integrity: The Transform Map's coalesce configuration and Dot Walking expressions ensure that all reference fields are correctly resolved, maintaining the relational integrity of the ServiceNow database.

Section 3 — Data Flow Design

3. Data Flow Design

The data flow design specifies the precise sequence in which data moves through the system from its point of origin in the external spreadsheet application to its final presentation within the ServiceNow Incident form. Understanding the data flow is essential for ensuring that each component in the architecture is configured correctly and that the dependencies between components are respected during implementation.

The data flow is linear and unidirectional: data originates externally, enters ServiceNow through the Import Sets module, is staged and transformed, stored permanently in the User table, and is subsequently consumed by the Incident Management module. The following step descriptions detail each stage of this flow.

3.1 Data Flow Steps

STEP
01
External Source

Employee Data Preparation

Employee records are created and organised in Google Sheets or Microsoft Excel. The dataset is structured with five defined column headers: Name, Email, Department, Manager, and Manager Email. Each row represents one employee record. The file is validated for completeness and formatting accuracy, then exported in XLSX or CSV format.

STEP
02
Source → ServiceNow

Data Upload via Import Sets

The administrator logs into ServiceNow and navigates to the System Import Sets module. Using the Load Data function, the exported employee data file is uploaded to ServiceNow. The platform reads the file and creates a new Import Set record with a corresponding staging table that mirrors the uploaded file's column structure.

STEP
03
Staging Layer

Data Staging in Import Set Table

The uploaded records are stored in the Import Set staging table. This staging table acts as a temporary repository holding the raw data in its original format before any transformation is applied. Administrators can inspect the staged records at this point to verify that the data has been loaded correctly and completely before proceeding.

STEP
04
Staging → sys_user

Data Transformation via Transform Map

The Transform Map is executed against the Import Set staging table. For each record, the Transform Map reads the source field values, applies the defined field mappings, resolves reference fields such as Department and Manager to their correct sys_id values, and evaluates Dot Walking expressions. The transformed record is then inserted or used to update the corresponding entry in the sys_user target table.

STEP
05
sys_user Table

Permanent Storage in User Table

The transformed employee records are permanently stored in the ServiceNow User table (sys_user). Each record contains correctly populated values for all five required fields, with reference fields properly resolved. The coalesce setting on the Email field ensures that repeated import operations update existing records rather than creating duplicates.

STEP
06
sys_user → Incident

Incident Assignment and Data Consumption

When a new incident is created and an employee is selected in the Assigned To field, the Incident table retrieves the reference to the corresponding sys_user record. The Dot Walking expressions configured in the Incident form layout then traverse this reference to automatically retrieve and display the employee's Department, Email, Manager, and Manager Email within the incident record.

STEP
07
Reference Traversal

Automatic Retrieval via Dot Walking

Dot Walking expressions on the Incident form dynamically follow the chain of reference field relationships from the Incident table through to the sys_user table and its related tables. The retrieved values are displayed in the Incident form in real time when an employee is selected, with no manual input required from the incident agent.

Section 4 — Database Design

4. Database Design

The database design defines the tables involved in the project, the fields they contain that are relevant to the import and incident management workflow, and the relationships between those tables. ServiceNow uses a relational database model where tables are interconnected through reference fields. Understanding these relationships is critical to designing the field mappings and Dot Walking expressions correctly, as incorrect assumptions about the relational structure will result in failed reference resolutions and missing data in the target records.

4.1 Table Descriptions

Table	Description and Role in the Project
Import Set Table (u_import_employee_data)	A temporary staging table automatically created by ServiceNow's Import Sets module when the employee data file is uploaded. This table mirrors the column structure of the source file and holds raw imported records before transformation. It is not a permanent production table and exists solely to facilitate the import and transformation process.
User Table (sys_user)	The primary production table that permanently stores all user records in ServiceNow and serves as the target table for the Transform Map.

Table	Description and Role in the Project
	Key fields relevant to this project include: name, email, department (reference to cmn_department), manager (self-referencing to another sys_user record), and the derived manager email field accessed via Dot Walking.
Department Table (cmn_department)	A reference table storing all department records in ServiceNow. The department field in sys_user is a reference field that points to records in this table. During the Transform Map execution, the department name string from the import file is resolved to the corresponding sys_id in cmn_department. If a matching department does not exist in this table, the reference will not resolve correctly.
Incident Table (incident)	Stores all incident records in ServiceNow. Extends the Task table and inherits its fields, including Priority. Consumes employee data from sys_user through the assigned_to and caller_id reference fields. Dot Walking expressions on the Incident form traverse these references to automatically display employee department and manager information.
Task Table (task)	The parent table from which the Incident table inherits common fields, including Priority, State, and Assignment Group. The Priority field is defined in the Task table's data dictionary and inherited by the Incident table. The Dictionary Override in this project targets the priority field at the Incident level only, overriding the inherited default without modifying the parent table definition.

4.2 Table Relationship Reference

Source Table	Relationship	Target Table	Fields Involved
`u_import_employee_data`	**transforms into →**	`sys_user`	`All five fields (Name, Email, Department, Manager, Manager Email)`
`sys_user`	**referenced by →**	`incident`	`assigned_to (sys_id), caller_id (sys_id)`
`sys_user`	**self-references →**	`sys_user`	`manager field (references another sys_user record)`

Source Table	Relationship	Target Table	Fields Involved
cmn_department	referenced by →	sys_user	department field (sys_id of department record)
task	extended by →	incident	All inherited fields including priority
incident	overridden by →	Dictionary	priority field (default value: 4 changed to 3)

4.3 Relationship Design Rationale

The relational design of the ServiceNow database means that the success of the import automation depends on the correct resolution of reference fields during the transformation process. The department field in sys_user stores the sys_id of the corresponding department record rather than the department name as a plain text string. Similarly, the manager field stores the sys_id of the manager's user record in sys_user.

This relational structure is precisely what enables Dot Walking to function effectively. Because the manager field is a reference to another sys_user record, Dot Walking can follow this reference and retrieve any field from that referenced record simply by constructing the appropriate expression. The design explicitly accounts for these relational dependencies by confirming that all referenced records exist in their respective tables before the import transformation is executed.

Section 5 — Field Design

5. Field Design

The field design specifies each data attribute involved in the import process, its data type, the manner in which it is mapped between the source and target tables, and any special handling

requirements. A precise field design is the foundation of an accurate Transform Map configuration and directly determines the quality of the employee records stored in sys_user after transformation.

5.1 Field Mapping Specification

Source Field	Data Type	Target Field (sys_user)	Mapping Method	Special Handling
`Name`	String	`name`	Auto-Mapped	Mapped directly as a plain text string to the full name field in sys_user. No special handling required.
`Email`	String	`email`	Auto-Mapped	Designated as the coalesce field. Used as the unique record identifier to prevent duplicate entries during repeated import operations.
`Department`	String	`department`	Auto / Manual	Reference field. The department name string is resolved to the corresponding sys_id in the cmn_department table during transformation.
`Manager`	String	`manager`	Manual	Self-referencing reference field. The manager's name is resolved to the sys_id of the corresponding sys_user record during transformation.
`Manager Email`	String	`manager.email`	**Manual (Dot Walk)**	Not a direct target field. Populated via the Dot Walk

Source Field	Data Type	Target Field (sys_user)	Mapping Method	Special Handling
				expression manager.email, traversing the manager reference to retrieve the email of the referenced manager record.

5.2 Coalesce Field Design

The designation of the Email field as the coalesce key is a critical design decision that determines how the Transform Map handles records during repeated import operations. When a coalesce field is specified, ServiceNow uses its value to search the target table for an existing matching record before deciding whether to insert a new record or update an existing one.

Coalesce Logic: Email Field as Unique Identifier

When a record with the same email address already exists in sys_user:

- The existing record is updated with the new field values from the import file.
- No duplicate record is created in sys_user.

When no matching email address is found in sys_user:

- A new user record is inserted into sys_user.
- All field values from the import file are populated in the new record.

Section 6 — Form Design

6. Form Design

The form design defines the customisations applied to the ServiceNow Incident form to enable it to display employee-related information automatically through Dot Walking. By default, the Incident form does not display the department or manager details of the assigned-to user. These fields must be explicitly added to the form layout using Dot Walking expressions so that they appear and auto-populate when an employee is selected in the Assigned To field.

The form design is guided by two principles: relevance and automation. Only fields that provide meaningful value to the incident handling workflow are added, and all added fields are configured to populate automatically through Dot Walking. No manual input is required from the incident agent once the Assigned To field is selected.

6.1 Dot Walking Field Specifications

Dot Walk Expression	Display Label on Form	Purpose
`assigned_to.department`	**Assigned To: Department**	Displays the name of the department to which the assigned employee belongs. Enables agents to identify the responsible business unit for incident routing and escalation.
`assigned_to.email`	**Assigned To: Email**	Displays the official email address of the assigned employee. Provides direct contact information without requiring the agent to navigate to the employee's user profile separately.
`assigned_to.manager`	**Assigned To: Manager**	Displays the full name of the direct line manager of the assigned employee. Enables agents to identify the correct escalation contact without additional lookup.
`assigned_to.manager.email`	**Assigned To: Manager Email**	A two-level Dot Walk that displays the email address of the assigned employee's manager. Enables direct manager contact for escalation without additional research by the agent.

6.2 Form Layout Design Decisions

- The Dot Walking fields are grouped together in a dedicated section of the Incident form, positioned immediately below the Assigned To field to which they are logically related, reinforcing the visual association between the reference field and its derived display values.
- All four Dot Walking fields are configured as read-only display fields, as their values are derived automatically from referenced records and must not be manually editable by incident agents.
- The fields are assigned descriptive display labels that clearly communicate to agents the source and meaning of the displayed information (e.g., Assigned To: Department rather than simply Department).
- The form layout is accessed through the Form Layout configuration tool via right-click Configure, which allows modifications to be made without impacting the underlying table schema or other views of the Incident form.

Section 7 — Priority Configuration Design

7. Priority Configuration Design

In ServiceNow's standard data model, the Priority field is defined in the Task table's data dictionary and inherited by all tables that extend Task, including the Incident table. The Task table defines a default priority value of 4 (Low), meaning that all newly created incident records receive a default priority of Low unless the agent manually selects a different value during incident creation.

The design decision for this project is to customise the default priority for incident records to 3 (Moderate) using ServiceNow's Dictionary Override mechanism. This decision is driven by the recognition that incidents represent unplanned disruptions to service, and a default priority of Low

understates the urgency typically associated with incident creation. By setting the default to Moderate, the design ensures that newly created incidents receive a baseline priority that better reflects the organisation's expected response standard, while still permitting agents to adjust priority as required by individual incident circumstances.

7.1 Priority Scale Reference

Priority Value	Priority Label	Applies To	Design Decision
1	Critical	All task-based tables	Inherited from Task — no override required.
2	High	All task-based tables	Inherited from Task — no override required.
3	**Moderate**	**Incident table only**	**SELECTED — Dictionary Override applied for all new incidents.**
4	Low	Task (original default)	System default before override. Replaced for Incident table.
5	Planning	All task-based tables	Inherited from Task — not applicable to incidents.

7.2 Dictionary Override Design Rationale

Design Rationale: Priority Default Override via Dictionary Override

- The Dictionary Override is applied at the Incident table level only, ensuring that other tables inheriting from Task (such as Problem and Change Request) are not affected by this customisation.
- The override modifies only the default value attribute of the Priority field, leaving all other attributes (data type, mandatory status, choice list values) unchanged.

- The override does not prevent agents from manually adjusting the priority of an incident during creation or editing; it only changes the pre-selected default value shown when a new incident form is opened.
- Using Dictionary Override rather than directly editing the Task table dictionary entry preserves the base system configuration and adheres to ServiceNow's best practice of avoiding direct modification of out-of-the-box table definitions.

Section 8 — Process Workflow Design

8. Process Workflow Design

The process workflow design describes the end-to-end sequence of activities that the implementation team will follow to build and configure the complete solution. The workflow is structured as an ordered series of discrete steps, each of which produces a specific output that serves as the prerequisite input for the step that follows. The workflow encompasses all activities from initial data preparation through to final configuration verification.

8.1 Implementation Workflow

Workflow Step	Activity Description and Expected Output
Step 1 Prepare Employee Data File	Create the employee dataset in Google Sheets or Microsoft Excel using the five defined column headers: Name, Email, Department, Manager, and Manager Email. Populate the dataset with representative employee records. Validate all entries for completeness, formatting consistency, and uniqueness of email addresses. Export the completed file in XLSX or CSV format. Expected Output: A validated, correctly formatted employee data file ready for upload to ServiceNow.
Step 2 Import File into ServiceNow	Log into ServiceNow as an administrator. Navigate to System Import Sets > Load Data. Create a new Import Set table and upload the

Workflow Step	Activity Description and Expected Output
	employee data file. Submit the form to initiate the data loading process. Verify that all records and columns are correctly loaded in the staging table. Expected Output: A fully populated Import Set staging table containing all employee records from the source file.
Step 3 Create Transform Map	Navigate to System Import Sets > Transform Maps. Create a new Transform Map record. Set the Source Table to the Import Set staging table and the Target Table to User [sys_user]. Save the record to generate the Field Maps related list. Expected Output: A saved Transform Map with source and target tables correctly configured.
Step 4 Map Fields Correctly	Use Auto Map Matching Fields to generate automatic mappings for matching fields. Manually create mappings for Manager and Manager Email. Enable the Coalesce checkbox on the Email field mapping. Enter the Dot Walking expression manager.email for the Manager Email field map. Review all mappings before proceeding. Expected Output: A complete, accurate set of field mappings covering all five required fields.
Step 5 Transform Data to User Table	Return to the Import Set and execute the Transform operation using the configured Transform Map. Monitor the transformation log for errors. Navigate to User Administration > Users to verify that all employee records are correctly created in sys_user with all required fields populated. Expected Output: Fully and accurately populated employee user records in the ServiceNow sys_user table.
Step 6 Configure Incident Form with Dot Walking	Open the Incident form and access the Form Layout configuration. Add the four Dot Walking fields: assigned_to.department, assigned_to.email, assigned_to.manager, and assigned_to.manager.email. Save the form layout. Create a test incident, assign it to an imported employee, and verify that all four fields auto-populate correctly. Expected Output: A customised Incident form that automatically displays employee department and manager information.
Step 7 Apply Dictionary Override for Priority	Navigate to System Definition > Dictionary. Locate the Priority field on the Incident table. Create a Dictionary Override entry setting the default value to 3 (Moderate). Save the override and create a new incident to confirm that the Priority field defaults to 3 — Moderate. Expected Output: Incident records that default to Priority 3 — Moderate upon creation.

8.2 Workflow Dependencies

The following sequential dependencies must be observed during implementation to ensure that each step has the necessary inputs from its predecessor:

1. Step 1 must precede Step 2: The Import Set requires a source data file to populate the staging table.
2. Step 2 must precede Step 3: The Transform Map requires an existing Import Set staging table as its source.
3. Step 3 must precede Step 4: Field mappings are created within the context of a saved Transform Map record.
4. Step 4 must precede Step 5: The transformation applies the complete field mapping configuration; incomplete mappings produce missing or incorrect data in sys_user.
5. Step 5 must precede Step 6: Dot Walking fields on the Incident form depend on correctly populated employee records in sys_user for verification purposes.
6. Steps 6 and 7 are independent of each other and may be performed in either order, but both must be completed before final validation.

Section 9 — Design Validation

9. Design Validation

Design validation is the process of critically evaluating the complete design specification against the project's requirements and success criteria to confirm that the proposed solution, if implemented as designed, will deliver the expected outcomes. Validation at the design stage is far more cost-effective than discovering design flaws during or after implementation, and it provides an important quality gate before development work begins.

The design validation for this project examines four key dimensions of system quality: data accuracy and storage integrity, automation of information retrieval, reduction of manual data entry, and improvement in incident handling efficiency. For each dimension, the design is assessed against the specific mechanisms and configurations that have been incorporated to achieve the desired outcome.

9.1 Validation Against Quality Dimensions

Quality Dimension	Design Features Addressing This Dimension	Validation Conclusion
Accurate Employee Data Storage	Import Set staging layer enables data validation before production records are created. Transform Map field mappings ensure correct field-to-field correspondence. Coalesce configuration on the Email field prevents duplicate records during repeated imports.	The design ensures that data is validated before commitment to sys_user and that all relational fields are correctly resolved. The staged import approach mitigates the risk of inaccurate or incomplete records being stored in production.
Automated Information Retrieval	Dot Walking expressions on the Incident form (assigned_to.department, assigned_to.email, assigned_to.manager, assigned_to.manager.email) retrieve related employee data dynamically from referenced records in sys_user and related tables.	The design fully automates the retrieval and display of employee attribute data within the Incident form. No manual data entry is required from agents once the Assigned To field is populated with an imported employee.
Reduced Manual Data Entry	The Transform Map automates the mapping and transformation of all employee records, eliminating the need for administrators to manually create or update user records in sys_user. The coalesce	The design eliminates manual data entry for both the initial import and for all subsequent repeated refresh operations. The overall administrative burden of maintaining employee records in ServiceNow is substantially reduced.

Quality Dimension	Design Features Addressing This Dimension	Validation Conclusion
	design ensures existing records are updated efficiently during refresh cycles.	
Faster Incident Handling	Dot Walking fields provide agents with immediate access to department and manager information directly within the Incident form without requiring additional navigation. Dictionary Override ensures all new incidents default to Moderate priority, reducing manual priority selection.	The design materially reduces the time agents spend gathering employee context information for incident routing and escalation. The priority default design further reduces the frequency of manual priority adjustments for standard incident records.

9.2 Design Completeness Checklist

The following checklist confirms that all required design elements have been fully addressed in this document and that the design specification is complete and ready to serve as the authoritative blueprint for the implementation phase:

1. System architecture has been defined across all seven layers, with each layer's role clearly described.
2. Data flow has been mapped across all seven steps, from external source preparation through to Incident form display.
3. Database design has been fully documented, including all relevant tables, their fields, relationships, and the rationale for the relational design approach.
4. Field design has been completely specified, covering data types, mapping methods, and special handling requirements for all five required fields.
5. Form design has been defined with all four Dot Walking expressions specified and the form layout design decisions described.

6. Priority configuration design has been documented with the complete priority scale reference and detailed rationale for the Dictionary Override approach.
7. Process workflow has been specified as a seven-step implementation sequence with all inter-step dependencies mapped.
8. All design decisions have been validated against the four project quality dimensions and assessed as adequate to meet the project's success criteria.

9.3 Design Readiness Statement

Upon completion of this design document, the project is assessed as fully ready to proceed to the Implementation Phase. All components of the solution have been designed in sufficient detail to enable implementation to proceed without requiring further design decisions during the development process. The design has been validated against the project's requirements and quality dimensions, and no outstanding design issues or unresolved questions have been identified.

The implementation team should use this document as the authoritative reference for all configuration activities during the development phase. Any deviations from the design specification encountered during implementation — for example, due to environmental constraints or platform version differences — should be documented and reviewed against the project's requirements and acceptance criteria before proceeding.

— End of Project Design Phase Document —

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